

WASP-36b

R-1776

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-36_161227 · created 2026-08-02T22:43:08+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457749.8188 +/- 0.0022 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.116 +/- 0.048
Transit depth (Rp/Rs)^2	1.35 +/- 1.11 [%]
Orbital Inclination (inc)	80.81 +/- 2.82
Airmass coefficient 1 (a1)	1.07 +/- 0.53
Airmass coefficient 2 (a2)	-0.019 +/- 0.049
Scatter in the residuals of the lightcurve fit is	51.84 %
Best Comparison Star	None
Optimal Aperture	3.0
Optimal Annulus	6.5
Transit Duration (day)	0.044 +/- 0.031

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.116 ± 0.048 vs 0.1368 (deviation 0.43σ)
Duration fit vs published	0.044 d vs 0.0758 d (deviation 1.03σ)
Mid-time O–C	-5.1 min
Depth SNR	2.4
Residual scatter	51.84 %

Light curve

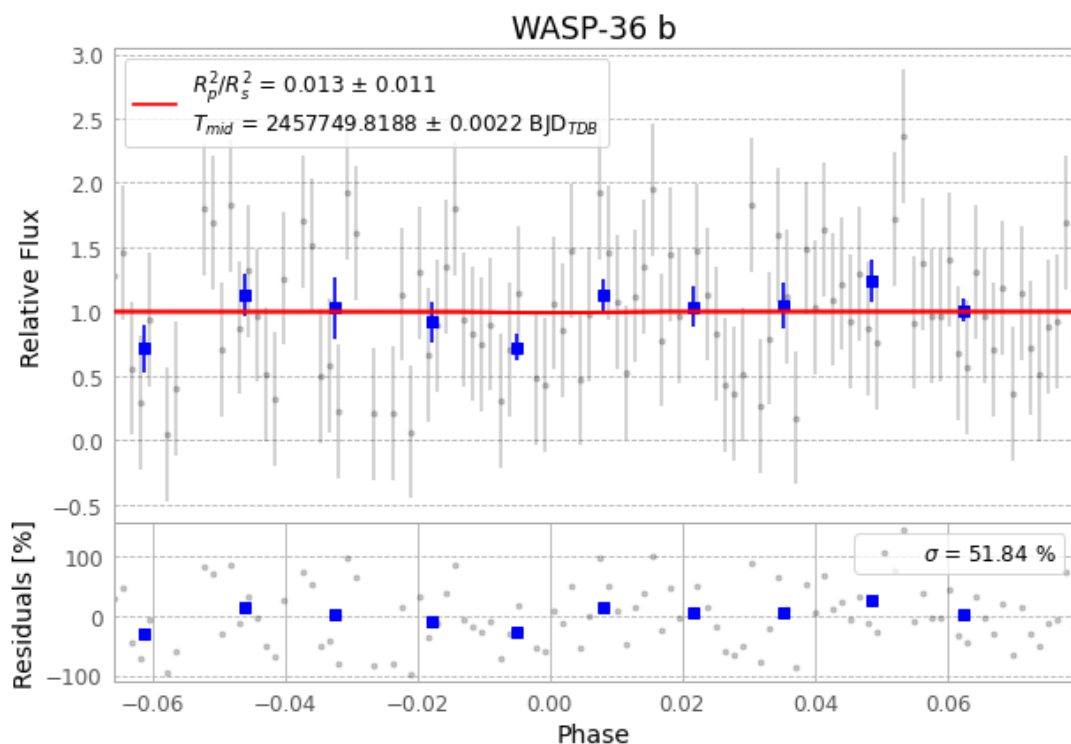


Figure 1 — FinalLightCurve_WASP-36 b_2016-12-26.png (SHA-256 3c9b02e6acb18838...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [160, 401], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 04ddc7556645961c...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

108 FITS frames (71 MB), WASP-36161227050612.FITS → WASP-36161227102712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-36-161227.log (3b7fa313e189...), FinalLightCurve_WASP-36 b_2016-12-26.png (3c9b02e6acb1...), FinalLightCurve_WASP-36 b_2016-12-26.csv (0ffefdb0f8a2...)

Compute resources

Wall time 170.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 22:43Z.